A4 Unity Catalog

Three-level namespace, grants, row filters and column masks.

A4-L1 The three-level namespace, one catalog per environment

The strong version reads exactly the table its config names, whatever the session's current catalog is.

```
research_prod.research_silver.nat_respondents
```

```
def table_name(catalog: str, layer: str, tracker: str) -> str:
    suffix = {"bronze": "raw", "silver": "respondents", "gold": "metrics"}[layer]
    return f"{catalog}.research_{layer}.{tracker}_{suffix}"
```

Unity Catalog name format	catalog.schema.table, for example research_prod.research_silver.nat_respondents.
What does each level hold in this project?	Catalog: environment. Schema: layer or governance. Table: tracker at a grain.
Why never write a two-part table name in pipeline code?	It resolves against the session's current catalog, which varies by cluster, notebook and job.
Where do table names come from?	One function built from config, such as table_name(catalog, layer, tracker).

A4-L2 Grant to groups, grant the least, grant at the right level

Three rules: groups not users, schemas not tables, SELECT for people and MODIFY for pipelines.

```
GRANT USE CATALOG ON CATALOG research_prod TO `research_analysts`;
GRANT USE SCHEMA, SELECT ON SCHEMA research_prod.research_gold TO
  `research_analysts`;
SHOW GRANTS ON SCHEMA research_prod.research_gold;
```

Privileges needed to read a table	USE CATALOG on the catalog, USE SCHEMA on the schema, SELECT on the table or schema.
Why grant SELECT on the schema rather than the table?	Schema grants are inherited by every current and future table in it.
Who gets MODIFY on production silver and gold?	The pipeline service principal. People get SELECT.
Why never grant to a user account?	User grants outlive the reason for them. Group membership is managed in one place.

A4-L3 Row filters and column masks in Unity Catalog

The strong script runs on Databricks, names groups instead of people, and uses three-part names throughout.

```
CREATE FUNCTION research_prod.research_governance.respondent_filter(tracker_id
  STRING)
RETURN is_account_group_member('research_approved_analysts') OR tracker_id = 'TRK-
  PUBLIC';

ALTER TABLE research_prod.research_silver.nat_respondents
SET ROW FILTER research_prod.research_governance.respondent_filter ON (tracker_id);
```

```
CREATE FUNCTION research_prod.research_governance.mask_postcode(postcode STRING)
```

```
RETURN CASE WHEN is_account_group_member('pii_readers') THEN postcode ELSE '***' END;

ALTER TABLE research_prod.research_silver.nat_respondents
ALTER COLUMN postcode SET MASK research_prod.research_governance.mask_postcode;
```

Row filter in Databricks	A SQL UDF returning BOOLEAN, bound with ALTER TABLE t SET ROW FILTER f ON (col).
Column mask in Databricks	A SQL UDF taking the column value, bound with ALTER TABLE t ALTER COLUMN c SET MASK f.
is_account_group_member or is_member?	is_account_group_member: Unity Catalog groups are account-level. is_member checks workspace groups.
What is CREATE ROW ACCESS POLICY?	Snowflake syntax. Databricks rejects it. Reference defect 4.